

Exponentials & Polynomials



Laws of exponents

- 1 Simplify $x^5 \cdot x^3$.
 - 2 Simplify $\frac{y^9}{y^4}$.
 - 3 Simplify $(2x^3)^4$.
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Negative and fractional exponents

- 4 Simplify x^{-3} , writing your answer without a negative exponent.
 - 5 Simplify $16^{3/4}$.
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Adding, subtracting, and multiplying polynomials

- 6 Simplify $(3x^2 + 2x - 5) + (x^2 - 4x + 7)$.
 - 7 Expand $(x + 3)(x - 5)$.
 - 8 Expand $(2x - 1)^2$.
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Factoring polynomials

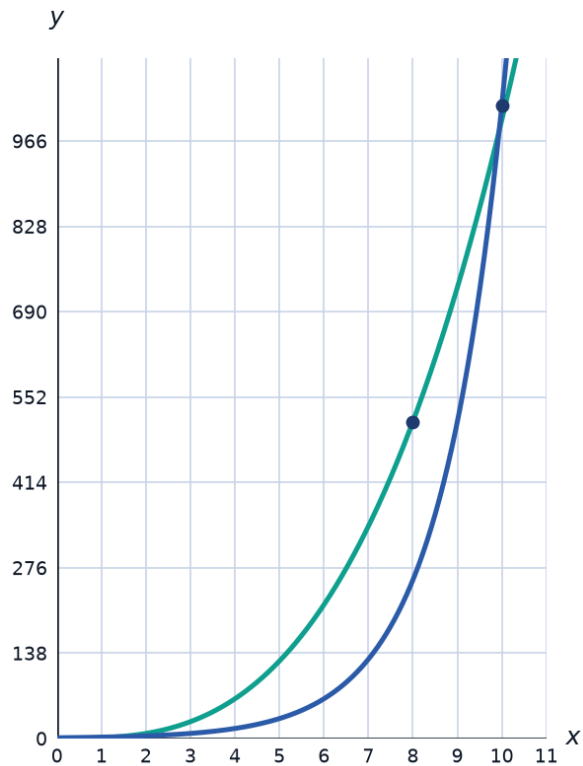
- 9 Factor $x^2 - 16$ completely.
 - 10 Factor $2x^2 + 8x$ completely.
 - 11 Factor $x^3 - 4x$ completely.
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Polynomial division and the remainder theorem

- 12 Find the remainder when $f(x) = x^3 - 2x^2 + x - 5$ is divided by $(x - 2)$, using the remainder theorem.
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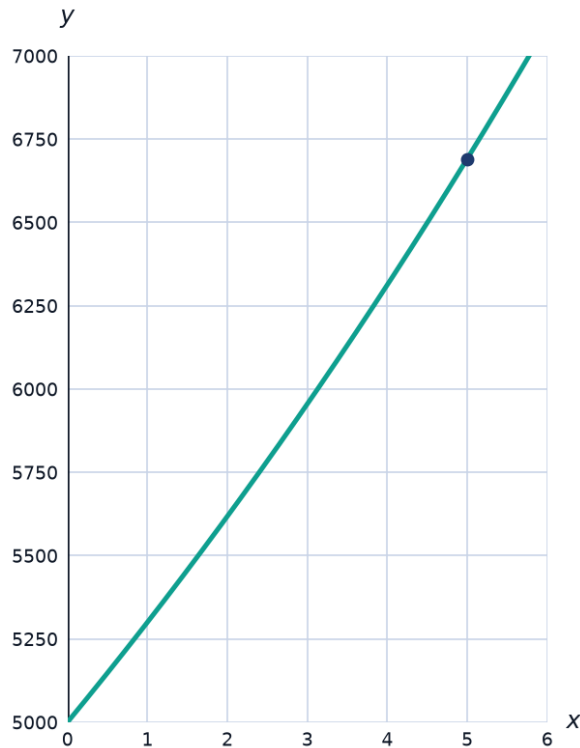
Exponential growth vs. polynomial growth

- 13 Two functions are given: $f(x) = x^3$ and $g(x) = 2^x$. Explain which function eventually grows faster as x increases, and estimate the crossover point using the values $f(10) = 1000$, $g(10) = 1024$, $f(8) = 512$, $g(8) = 256$.



Modeling with exponential functions

- 14** An investment of 5000 dollars grows according to $A(t) = 5000(1.06)^t$, where t is the number of years. Find the value of the investment after 5 years, to the nearest dollar, and state the annual growth rate as a percentage.



Applying the zero product property to polynomial equations

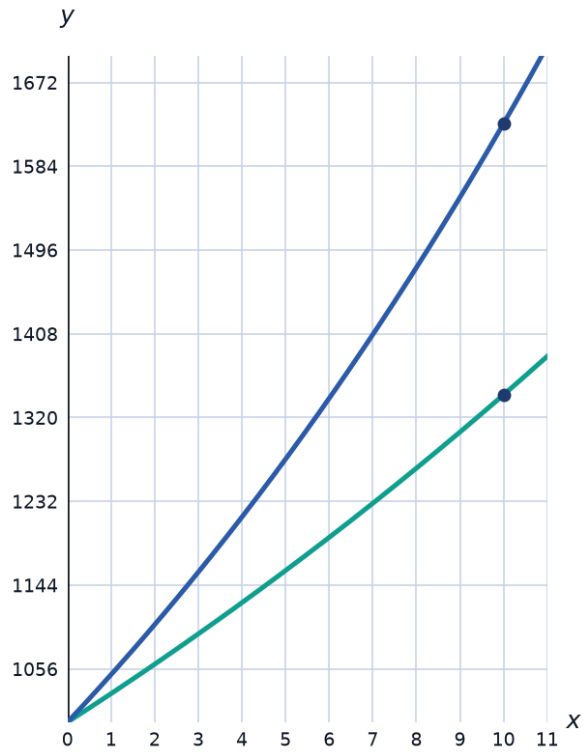
- 15** Solve $x^3 - 5x^2 + 6x = 0$ by factoring completely and applying the zero product property.

Multiplying binomials and trinomials

- 16** Expand $(x + 2)(x^2 - 3x + 4)$.
- 17** Expand $(2x - 3)(x + 5)$.

Comparing exponential models

- 18 Two savings accounts grow according to $A(t) = 1000(1.03)^t$ and $B(t) = 1000(1.05)^t$. Explain which account grows faster, and find the difference in value after 10 years, to the nearest dollar.



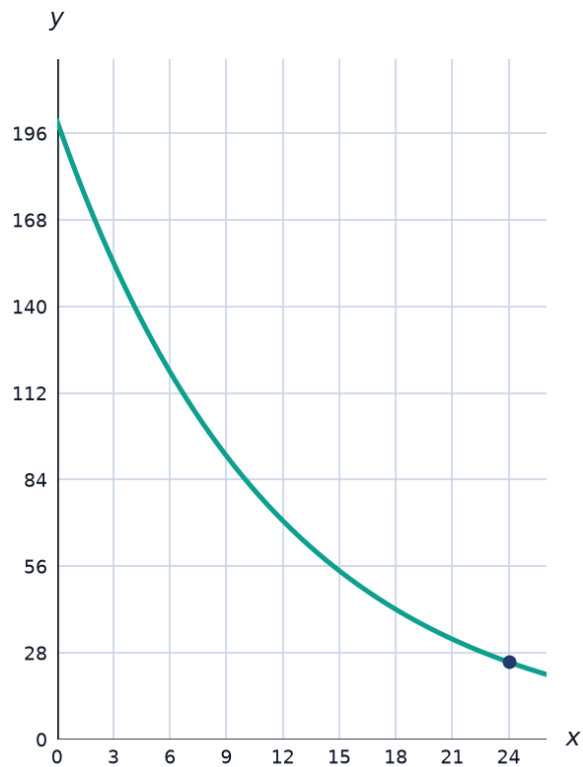
Simplifying expressions with multiple exponent rules

19 Simplify $\frac{(x^3y^2)^2}{x^4y}$.

20 Simplify $\left(\frac{2x^2}{y}\right)^3$.

Half-life and decay applications

- 21 A radioactive substance has a half-life of 8 years, starting with 200 grams. Write a model for the remaining amount after t years, and find the amount remaining after 24 years.



Multiple choice — easy 15

- 22 Simplify x^{12}/x^2 .

- a. x^{10}
- b. x^{12}
- c. x^{11}
- d. x^9

23 Simplify $x^6 \cdot x^5$.

a. x^{12}

b. x^{13}

c. x^{10}

d. x^{11}

24 Simplify $(x^4)^4$.

a. x^{15}

b. x^{16}

c. x^{17}

d. x^{18}

25 Simplify $x^9 \cdot x^7$.

a. x^{18}

b. x^{16}

c. x^{17}

d. x^{15}

26 Simplify x^{18}/x^9 .

a. x^{10}

b. x^9

c. x^{11}

d. x^8

27 Simplify x^{10}/x^2 .

a. x^9

b. x^8

c. x^7

d. x^{10}

28 Simplify $x^5 \cdot x^8$.

a. x^{14}

b. x^{15}

c. x^{13}

d. x^{12}

29 Simplify $x^{10} \cdot x^8$.

a. x^{17}

b. x^{19}

c. x^{18}

d. x^{20}

30 Simplify $x^5 \cdot x^6$.

a. x^{10}

b. x^{12}

c. x^{11}

d. x^{13}

31 Simplify x^9/x^2 .

a. x^6

b. x^8

c. x^9

d. x^7

32 Simplify x^{13}/x^3 .

a. x^{12}

b. x^{10}

c. x^{11}

d. x^9

33 Simplify $x^{10} \cdot x^6$.

a. x^{16}

b. x^{18}

c. x^{15}

d. x^{17}

34 Simplify x^9/x^7 .

a. x^4

b. x^3

c. x^2

d. x^1

35 Simplify $(x^2)^5$.

a. x^{11}

b. x^{10}

c. x^9

d. x^{12}

36 Simplify $(x^4)^5$.

a. x^{20}

b. x^{22}

c. x^{21}

d. x^{19}

Multiple choice — medium 15

37 Expand $(4x - 7)(5x + 9)$.

a. $20x^2 + x + 63$

b. $20x^2 + 3x - 63$

c. $20x^2 + x - 63$

d. $20x^2 - x - 63$

38 Expand $(3x - 6)(2x + 5)$.

a. $6x^2 + 3x - 30$

b. $6x^2 + 3x + 30$

c. $6x^2 - 3x - 30$

d. $6x^2 + 5x - 30$

39 Expand $(x + 9)(5x + 5)$.

a. $5x^2 - 50x + 45$

b. $5x^2 + 50x - 45$

c. $5x^2 + 50x + 45$

d. $5x^2 + 52x + 45$

40 Expand $(3x + 2)(4x + 5)$.

a. $12x^2 - 23x + 10$

b. $12x^2 + 23x - 10$

c. $12x^2 + 23x + 10$

d. $12x^2 + 25x + 10$

41 Expand $(4x + 9)(4x + 8)$.

a. $16x^2 + 70x + 72$

b. $16x^2 - 68x + 72$

c. $16x^2 + 68x + 72$

d. $16x^2 + 68x - 72$

42 Expand $(4x + 9)(2x + 4)$.

a. $8x^2 + 34x + 36$

b. $8x^2 + 36x + 36$

c. $8x^2 + 34x - 36$

d. $8x^2 - 34x + 36$

43 Expand $(5x + 5)(3x - 2)$.

a. $15x^2 + 5x + 10$

b. $15x^2 + 7x - 10$

c. $15x^2 + 5x - 10$

d. $15x^2 - 5x - 10$

44 Expand $(5x + 6)(3x + 5)$.

a. $15x^2 + 45x + 30$

b. $15x^2 + 43x + 30$

c. $15x^2 + 43x - 30$

d. $15x^2 - 43x + 30$

45 Expand $(5x - 3)(2x - 1)$.

a. $10x^2 - 11x - 3$

b. $10x^2 + 11x + 3$

c. $10x^2 - 11x + 3$

d. $10x^2 - 9x + 3$

46 Expand $(5x + 9)(3x + 8)$.

a. $15x^2 + 69x + 72$

b. $15x^2 + 67x + 72$

c. $15x^2 - 67x + 72$

d. $15x^2 + 67x - 72$

47 Expand $(2x + 8)(4x - 3)$.

a. $8x^2 + 26x + 24$

b. $8x^2 + 28x - 24$

c. $8x^2 - 26x - 24$

d. $8x^2 + 26x - 24$

48 Expand $(3x - 3)(5x + 3)$.

a. $15x^2 - 6x - 9$

b. $15x^2 + 6x - 9$

c. $15x^2 - 4x - 9$

d. $15x^2 - 6x + 9$

49 Expand $(3x - 9)(x - 7)$.

a. $3x^2 - 30x - 63$

b. $3x^2 - 28x + 63$

c. $3x^2 - 30x + 63$

d. $3x^2 + 30x + 63$

50 Expand $(3x + 7)(2x + 4)$.

a. $6x^2 - 26x + 28$

b. $6x^2 + 26x + 28$

c. $6x^2 + 26x - 28$

d. $6x^2 + 28x + 28$

51 Expand $(2x + 9)(2x - 8)$.

- a. $4x^2 + 2x - 72$
- b. $4x^2 - 2x - 72$
- c. $4x^2 + 4x - 72$
- d. $4x^2 + 2x + 72$

Multiple choice — hard 15

52 Find the remainder when $f(x) = x^3 + 5x^2 + 2x + 3$ is divided by $(x - (-2))$, using the remainder theorem.

- a. 11
- b. 9
- c. 13
- d. -11

53 Find the remainder when $f(x) = x^3 - 6x^2 - 7x - 15$ is divided by $(x - (2))$, using the remainder theorem.

- a. -45
- b. 45
- c. -43
- d. -47

54 Find the remainder when $f(x) = x^3 + 3x^2 + 9$ is divided by $(x - (-4))$, using the remainder theorem.

- a. -5
- b. -7
- c. -9
- d. 7

- 55 Find the remainder when $f(x) = x^3 + 3x^2 + 6x + 14$ is divided by $(x - (-1))$, using the remainder theorem.
- a. 12
 - b. 10
 - c. -10
 - d. 8
- 56 Find the remainder when $f(x) = x^3 - 4x^2 - 4x - 15$ is divided by $(x - (-1))$, using the remainder theorem.
- a. -16
 - b. -14
 - c. -18
 - d. 16
- 57 An investment of 3000 dollars grows according to $A(t) = 3000(1.07)^t$, where t is the number of years. Find the value of the investment after 5 years, to the nearest dollar.
- a. 4408
 - b. 4008
 - c. 4208
 - d. 6311
- 58 An investment of 8000 dollars grows according to $A(t) = 8000(1.08)^t$, where t is the number of years. Find the value of the investment after 4 years, to the nearest dollar.
- a. 11084
 - b. 10884
 - c. 10684
 - d. 16326

- 59 An investment of 3000 dollars grows according to $A(t) = 3000(1.07)^t$, where t is the number of years. Find the value of the investment after 8 years, to the nearest dollar.
- a. 5155
 - b. 4955
 - c. 7732
 - d. 5355
- 60 An investment of 5000 dollars grows according to $A(t) = 5000(1.05)^t$, where t is the number of years. Find the value of the investment after 6 years, to the nearest dollar.
- a. 6700
 - b. 6900
 - c. 6500
 - d. 10051
- 61 An investment of 8000 dollars grows according to $A(t) = 8000(1.03)^t$, where t is the number of years. Find the value of the investment after 8 years, to the nearest dollar.
- a. 10334
 - b. 10134
 - c. 15201
 - d. 9934
- 62 Solve $x^3 + 3x^2 - 10x = 0$ by factoring completely. Excluding the solution $x = 0$, find the other two solutions as an ordered pair (smaller first).
- a. $(-5, 1)$
 - b. $(-4, 2)$
 - c. $(-5, 2)$
 - d. $(5, -2)$

- 63** Solve $x^3 + 11x^2 + 18x = 0$ by factoring completely. Excluding the solution $x = 0$, find the other two solutions as an ordered pair (smaller first).
- a. $(9, 2)$
 - b. $(-9, -2)$
 - c. $(-9, -3)$
 - d. $(-8, -2)$
- 64** Solve $x^3 + 3x^2 - 40x = 0$ by factoring completely. Excluding the solution $x = 0$, find the other two solutions as an ordered pair (smaller first).
- a. $(-8, 4)$
 - b. $(8, -5)$
 - c. $(-8, 5)$
 - d. $(-7, 5)$
- 65** Solve $x^3 + 7x^2 - 18x = 0$ by factoring completely. Excluding the solution $x = 0$, find the other two solutions as an ordered pair (smaller first).
- a. $(9, -2)$
 - b. $(-9, 1)$
 - c. $(-9, 2)$
 - d. $(-8, 2)$
- 66** Solve $x^3 - 15x^2 + 56x = 0$ by factoring completely. Excluding the solution $x = 0$, find the other two solutions as an ordered pair (smaller first).
- a. $(7, 8)$
 - b. $(7, 7)$
 - c. $(-7, -8)$
 - d. $(8, 8)$